Name: Yalezwa, Malaya, Cherish, Keisha **Date:** 2/10/26 **Class:** e4usa

MyDesign Portfolio

Element D Initial Template

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation *(D1)*

Design Idea #1: Rounded Scraper the width of the panini press

Date: 2/2/2026
Description: A 14 inch steel blade used to get rid of gunk and burnt on food, with plastic heat resistant handles on both sides.

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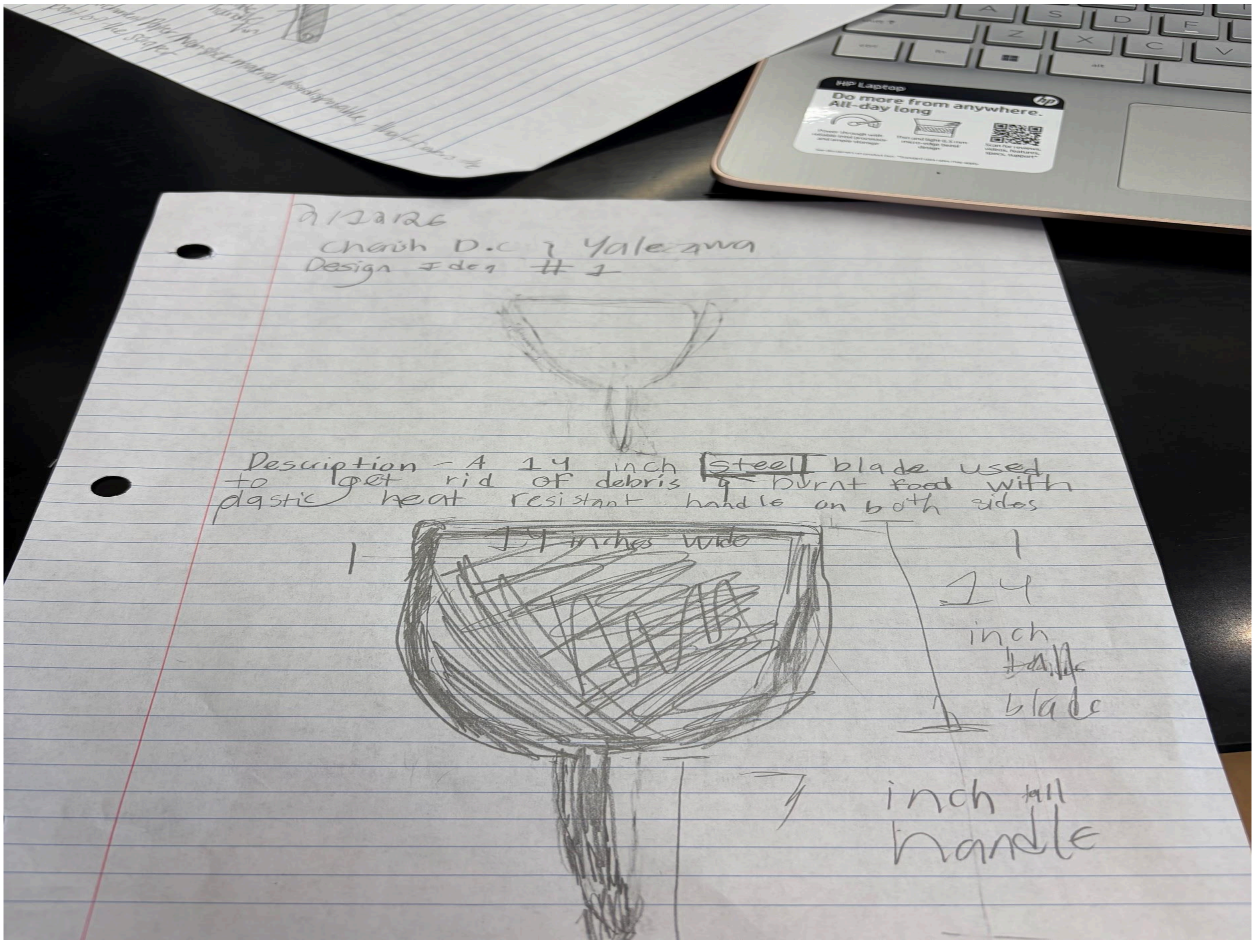

Design Idea #2: Scraper Covers (Non-Stick)

Date: 2/2/2026
Description: Plastic/Parchment Paper/Nonstick material, also disposable, that covers the blade part of the scraper.
Drawing that includes distinguishing information such as scale, dimensions, and materials:

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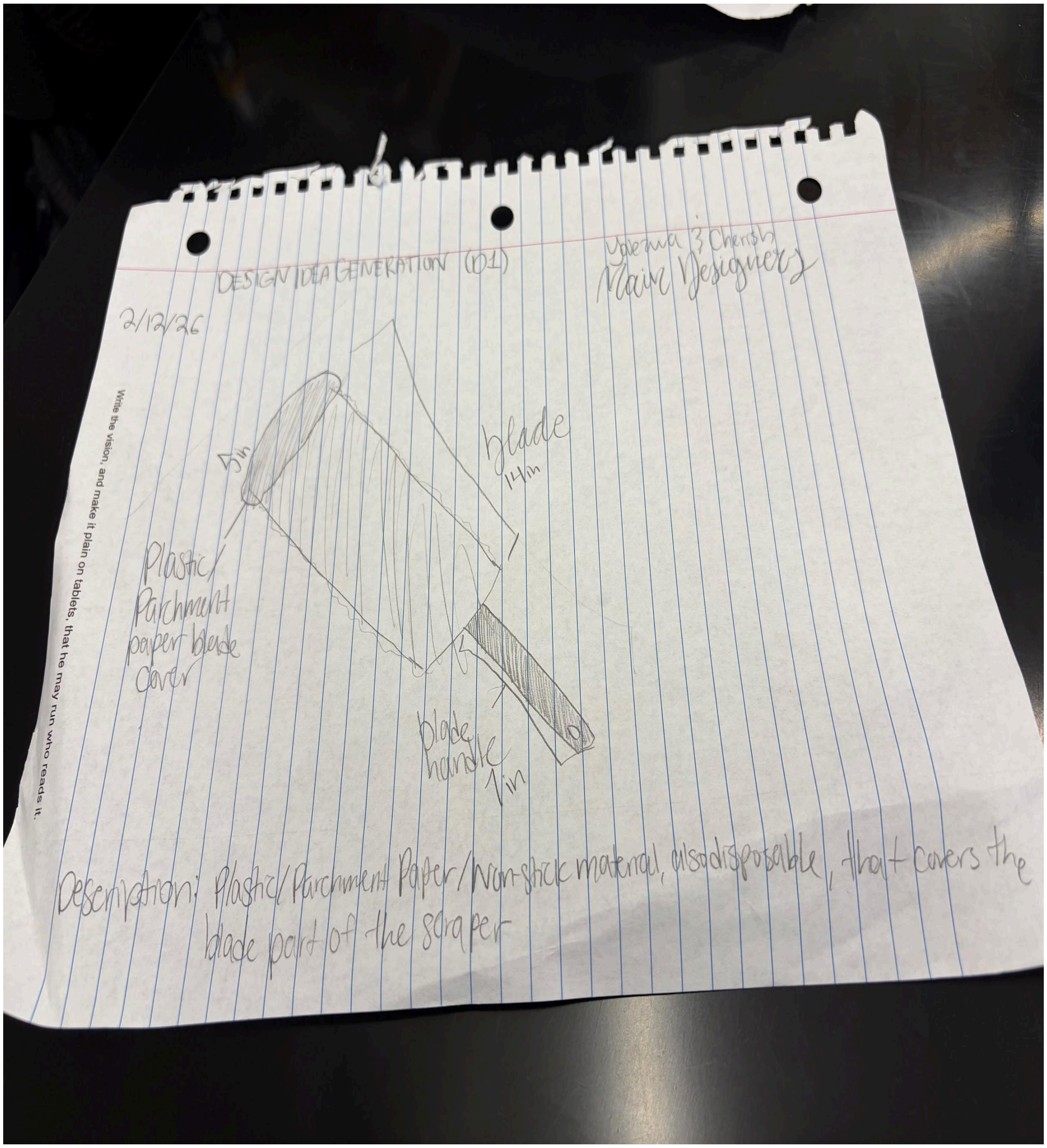

Design Idea #3: Large Dust Pan for scraps

Name: Yalezwa, Malaya, Cherish, Keisha **Date:** 2/10/26 **Class:** e4usa

Date: 2/2/2026

Description: A wider, taller dust pan to catch debris from the panini press that food won't fall from the gap to cause less mess.

Drawing that includes distinguishing information such as scale, dimensions, and materials:

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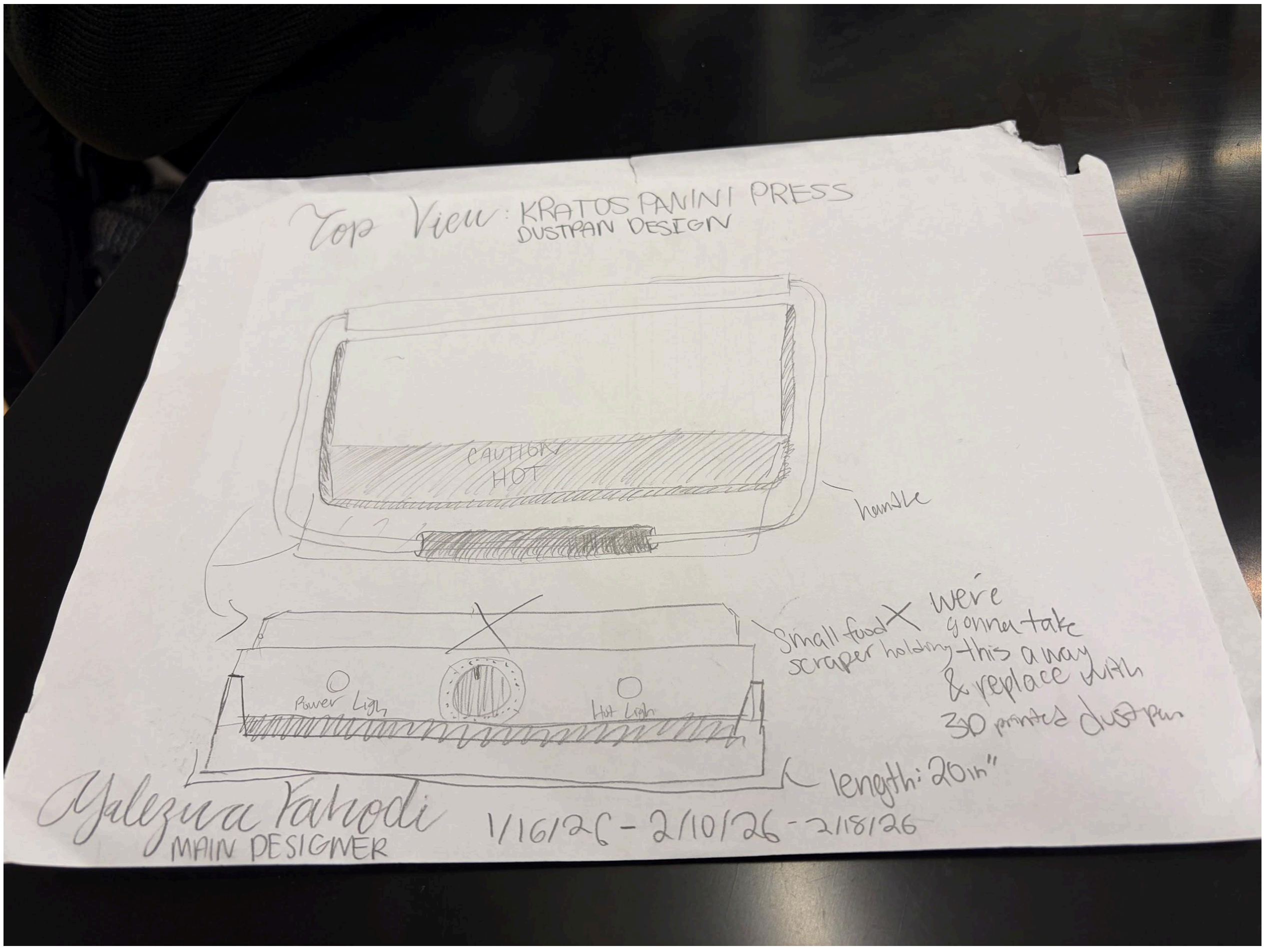

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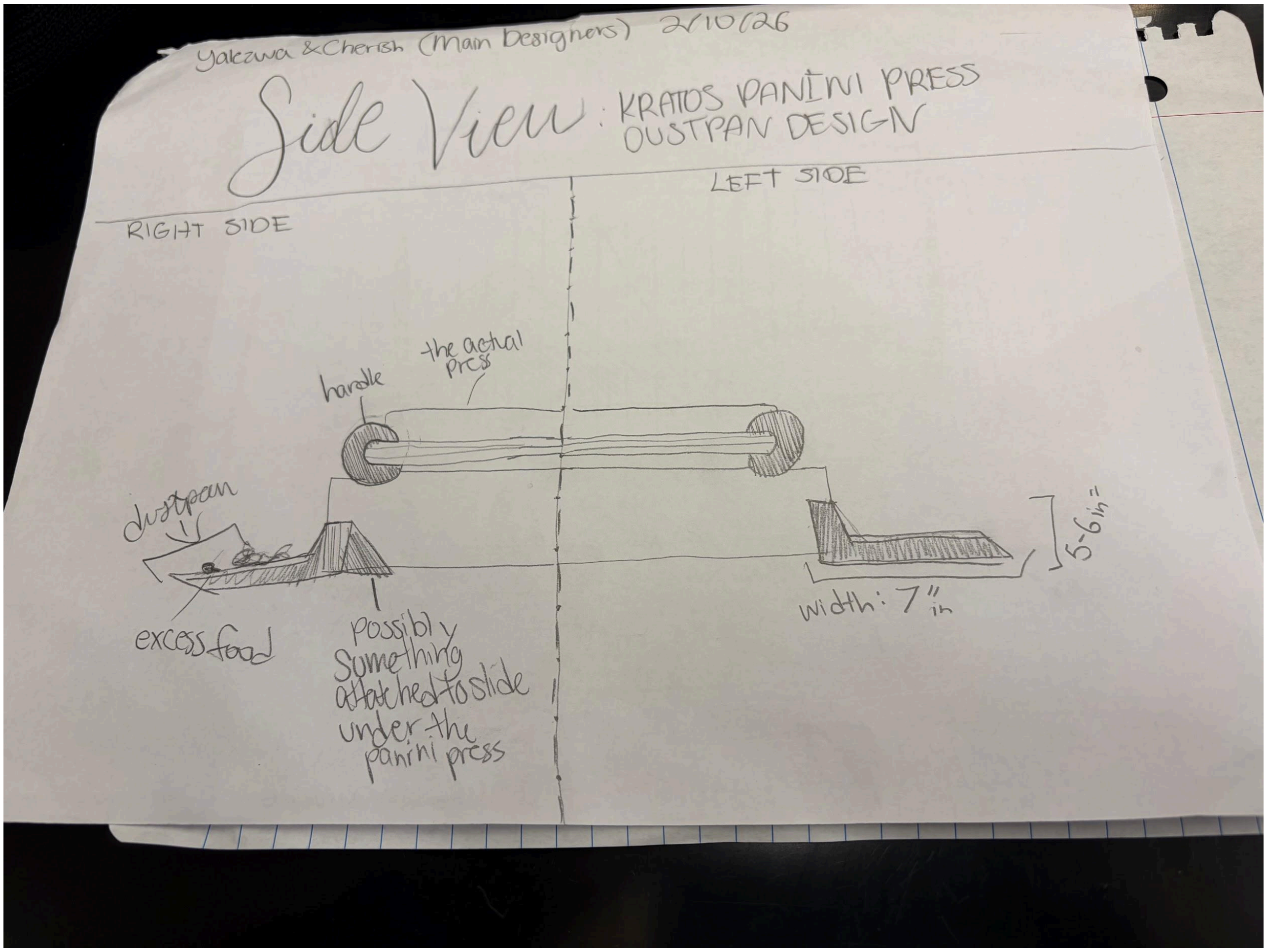


3d Design Idea Comparison and Selection *(D1, D2)*

Name: Yalezwa, Malaya, Cherish, Keisha **Date:** 2/10/26 **Class:** e4usa

	Weight 1-10	Scrap Covers	Rounded Scraper	Dustpan	Panini Press Covers
Heat Resistant	weight	**2**	**8**	**9**	**8**
Protect Panini Press	weight	**3**	**9**	**5**	**10**
Get rid of debris	weight	**2**	**10**	**9**	**3**
Efficent	weight	**8**	**9**	**10**	**10**
Easy to access	weight	**5**	**3**	**7**	**9**
score		**20**	**39**	**40**	**40**

Design Blueprint *(D4)*

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

-Recyclable Material

-Length of sides =13.5

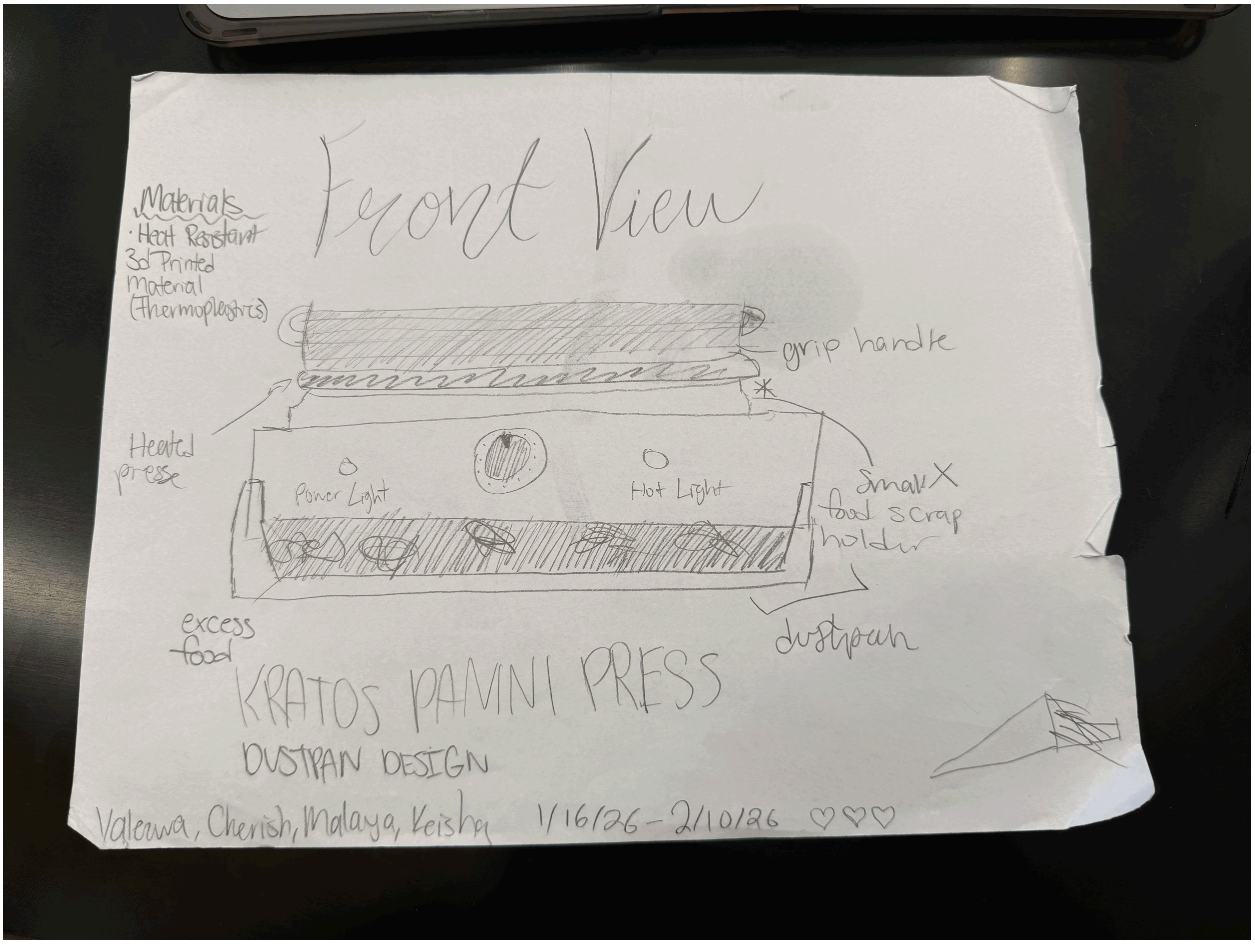
-Front =14

-Back =14

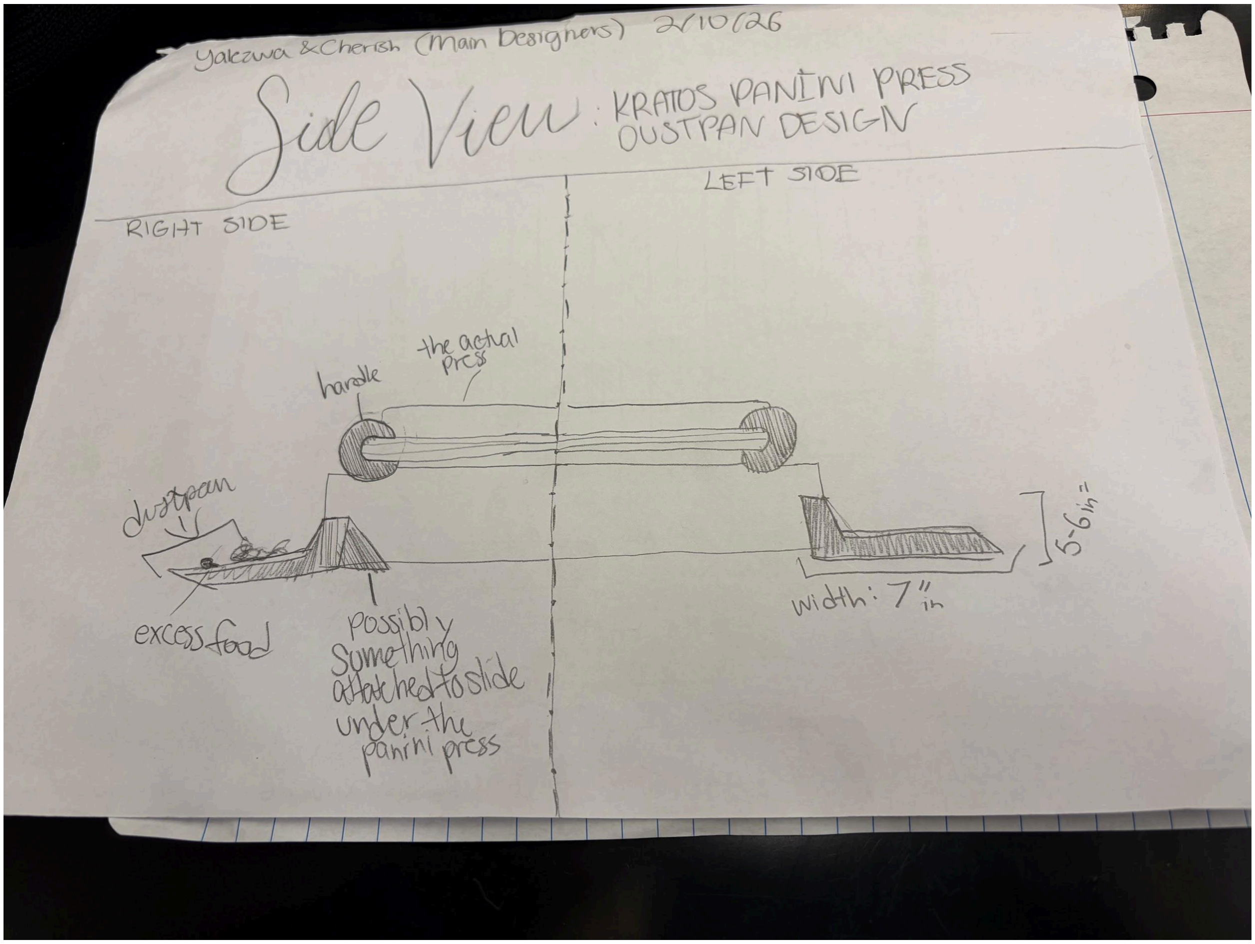
Design Blueprint #1

Name: Yalezwa, Malaya, Cherish, Keisha **Date:** 2/10/26 **Class:** e4usa

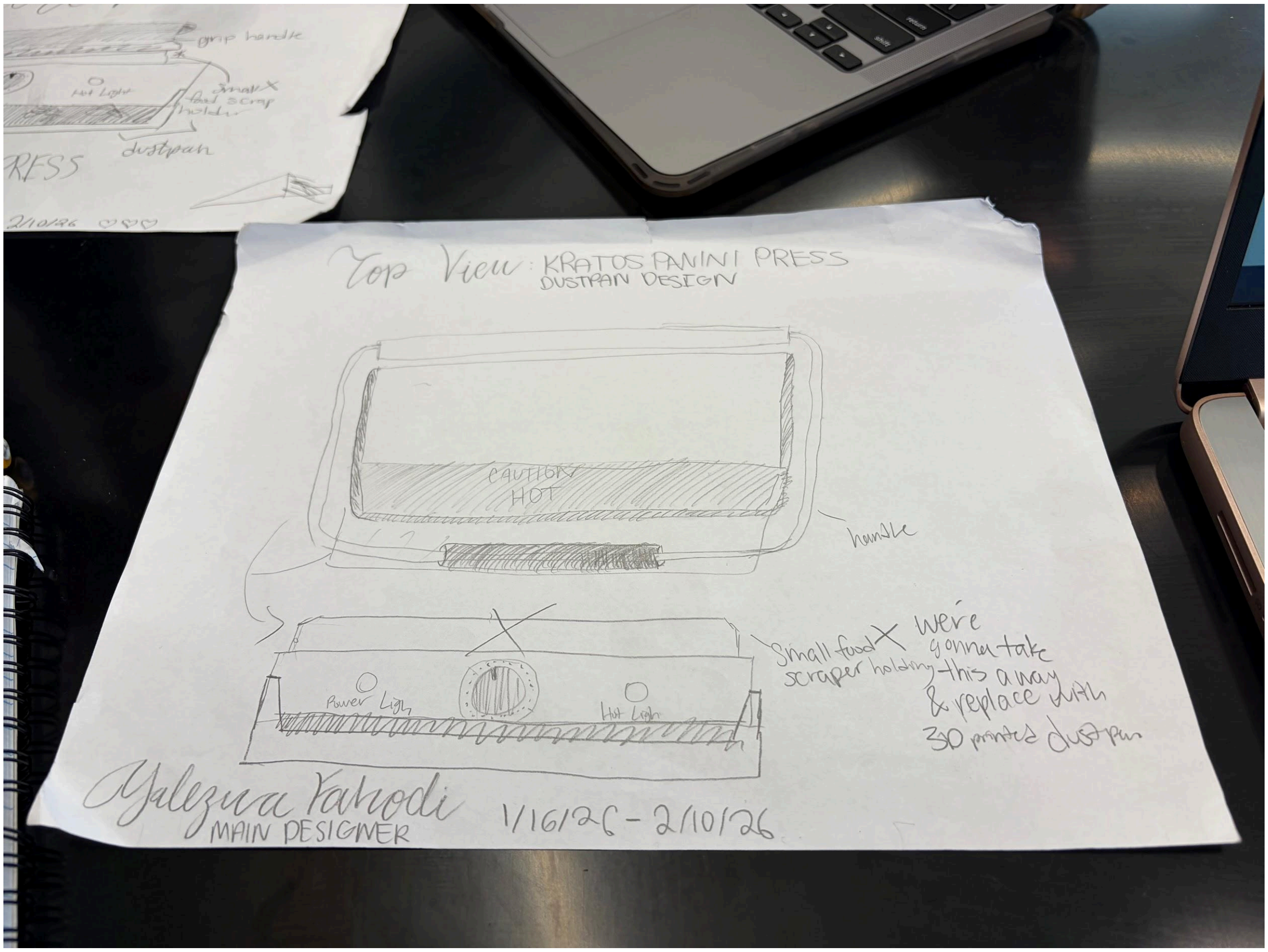
Date:2/10/26



Name: Yalezwa, Malaya, Cherish, Keisha Date: 2/10/26 Class: e4usa



Name: Yalezwa, Malaya, Cherish, Keisha **Date:** 2/10/26 **Class:** e4usa



Sketch: **KRATOS PANINI PRESS DUSTPAN DESIGN**

Manufacturing Methods:

Step 1: **3D Printing**

Step 2: **Ordering a cleaning prototype on amazon**

Step 3: **Building and Testing our prototype**

Supply List and Costs:

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Item	Cost each	# needed	Total
LR 3D Printer in Lab 4	$0	3	$0
3d Printed Material	$0	3	$0
Panini Press	$0	3	$0
Total Cost	$0	$0	$0

Plan of Action *(D3)*

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.

Phase 1:Research and Definition

- Market and Research
    - Evaluate other products similar to our panini press claiming tool.
- Design specifications
    - Finalize the requirements for cleaning: material safe, efficiently, and physically fit.

Phase 2: Concept Design and Material Choices

- STEM Analysis
    - Jot down the material science behind our choices of materials.

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- Sketching
    - Create a detailed 2D sketch that takes into account a clearance tolerance of 0.12in, due to thermal expansion.

Phase 3: Prototyping and Development

- Repeated building
    - Use heat-resistant silicone to construct the handle and cleaning scraper.
- Bench testing
    - Verify the mechanical advantage of the handle by measuring the force that is transferred to the surface of the panini press.

Phase 4: Expert Review and Testing

- Expert Concultations
    - Schedule interview with Mechanical Engineers or Materials scientists to review STEM justifications.
- Prototype Testing
    - Clean the heated panini press to test thermal conductivity safety and how efficient it removes residue.
- Revisions
    - Utilize constructive criticism based on the experts feedback.